BrickTrack — Launch Roadmap

Web App: Current State → Shippable Product

Target: End of May 2026 (earlier if possible)

Executive Summary

BrickTrack is a LEGO set tracking, rating, and discovery platform (Letterboxd for LEGO). The core product — search, collections, reviews, lists, and SEO — is fully functional. This roadmap covers the remaining work to go from development state to a polished, monetizable, user-ready web app with low ongoing maintenance costs.

Current state: ~80% feature-complete. Auth, search, collections, reviews, lists, themes, SEO all working. **Remaining work:** Polish, security hardening, real price data, email automation, design pass, and launch prep.

What's Already Built & Working

Area	Status
User registration & login (JWT)	Done
LEGO set database (10k+ sets via Rebrickable)	Done
Search with filters (theme, year, pieces, rating)	Done
Collections (owned + wishlist)	Done
Custom lists (create, share, reorder)	Done
Reviews & ratings (0.5-5 stars)	Done
Public user profiles	Done
Email signup capture	Done
Affiliate click tracking	Done
SEO (sitemaps, OG images, meta tags)	Done
GA4 analytics	Done
Mobile-responsive layout	Done
Dark mode support	Done
Backend test suite (16 test files)	Done
Render + Vercel deployment config	Done

Phase 1: Security & Auth Hardening (Week 1)

Priority: Critical — must ship before any real users

1.1 Password Reset Flow

- Add `POST /auth/forgot-password` endpoint (generates reset token, stores in DB)
- Add `POST /auth/reset-password` endpoint (validates token, updates password)
- Frontend `/forgot-password` and `/reset-password` pages
- Send reset email via transactional email service (see 3.1)
- Tokens expire after 1 hour

1.2 Email Verification

- Add `is_email_verified` field to users table
- Send verification email on registration
- Add `GET /auth/verify-email?token=...` endpoint
- Soft-gate: allow usage but show banner until verified
- Resend verification option in account settings

1.3 Admin Auth Protection

- Add `is_admin` field to users table
- Protect `/admin/*` endpoints with admin check middleware
- Create simple admin dashboard page (set refresh trigger, user count, email signups)

1.4 Rate Limiting

- Add rate limiting middleware (slowapi or custom)
- Auth endpoints: 5 requests/minute per IP
- API endpoints: 60 requests/minute per user
- Prevent abuse of search/suggest endpoints

1.5 Input Sanitization Audit

- Review all user-facing inputs (review text, list names, usernames)
- Ensure HTML/XSS prevention on all text fields
- Validate email format server-side

Phase 2: UI/UX Polish & Design Pass (Weeks 2-3)

Priority: High — directly impacts user retention and perception

2.1 Home Page Redesign

- Hero section with value proposition and CTA
- Trending/popular sets section with better visual hierarchy
- "Why BrickTrack?" feature highlights (track collection, compare prices, get alerts)
- Social proof section (total reviews, active users, sets tracked)

- Clean, modern aesthetic matching top consumer apps

2.2 Set Detail Page Final Polish

- Image gallery with zoom (if multiple images available)
- Better review UX (helpful votes, sort by date/rating)
- Related sets recommendations (same theme, similar piece count)
- Price history placeholder (show "Coming soon" until real data)
- Share button (copy link, social share)

2.3 Collection UX Improvements

- Collection stats dashboard (total pieces, total sets, theme distribution)
- Visual grid/list toggle for collection view
- "Add to collection" quick action from any set card
- Export collection as CSV/shareable link
- Import sets by number (bulk add)

2.4 Global Design System

- Consistent spacing, typography, and color palette across all pages
- Loading skeletons for all data-fetching states
- Empty states with helpful CTAs (no reviews yet, empty collection, etc.)
- Error states with retry actions
- Toast notifications for user actions (added to collection, review posted, etc.)
- Smooth page transitions

2.5 Mobile Polish Pass

- Test and fix all pages at 375px and 768px breakpoints
- Bottom navigation bar for mobile (Home, Search, Collection, Profile)
- Touch-friendly tap targets (minimum 44px)
- Swipe gestures for collection management

2.6 Onboarding Flow

- First-time user experience after signup
- "Pick your favorite themes" step (personalizes discover page)
- "Add your first sets" guided flow
- Tooltip hints for key features

Phase 3: Email & Notifications (Week 3)

Priority: High — drives retention and monetization

3.1 Transactional Email Setup

- Integrate email service (Resend, SendGrid, or AWS SES)
- Email templates: welcome, password reset, email verification
- Sender domain setup (SPF, DKIM, DMARC)

- Unsubscribe link in all emails

3.2 Deal Alerts (Email)

- "Notify me when this set goes on sale" button on set detail
- Store alert preferences per user per set
- Send email when price drops (once real price data exists)
- Weekly digest option: "Sets you're watching"

3.3 Retiring Soon Alerts

- "Alert me when sets I own/want are retiring" preference
- Weekly email: "These sets in your wishlist are retiring soon"
- Drives urgency and affiliate clicks

3.4 Review/Social Notifications

- Email when someone likes your review (future)
- Email when a list you follow gets updated (future)
- Keep these as Phase 2 stretch goals

Phase 4: Price Data Integration (Weeks 4-5)

Priority: High — core differentiator and monetization driver

4.1 LEGO.com MSRP Data

- Pull RRP/MSRP from Rebrickable set data (already available in API)
- Store retail_price and retail_currency on Set model
- Display MSRP on set detail page

4.2 Amazon Product Advertising API

- Set up Amazon Associates account + PA-API access
- Build backend service to query Amazon for LEGO set prices
- Map LEGO set numbers to Amazon ASINs (build mapping table)
- Cache prices with 6-hour TTL
- Display Amazon price + "Buy on Amazon" affiliate link

4.3 Price Comparison Infrastructure

- Replace hardcoded offers.py with database-backed offers table
- Schema: set_num, store, price, currency, url, in_stock, last_checked
- Periodic refresh job (cron/scheduled task)
- Support multiple stores (Amazon, LEGO.com, Walmart, Target)

4.4 Price Display Enhancements

- "Best price" badge on set cards in search/browse
- Price comparison table on set detail

- "Price dropped!" indicators
- Affiliate disclosure clearly visible

4.5 Future: Price History (Post-Launch)

- Track price changes over time in a history table
- Simple line chart on set detail page
- "All-time low" and "average price" stats

Phase 5: Performance & Reliability (Week 5)

Priority: Medium — ensures good experience at scale

5.1 Backend Performance

- Add Redis caching for hot endpoints (set detail, themes list, search)
- Database query optimization (check slow query log)
- Connection pooling tuning for PostgreSQL
- Gzip/Brotli compression on API responses

5.2 Frontend Performance

- Audit Core Web Vitals (LCP, CLS, FID)
- Lazy load below-fold components
- Image optimization audit (proper sizes, formats, lazy loading)
- Bundle size analysis and code splitting review
- Prefetch critical navigation paths

5.3 Error Monitoring

- Add Sentry (free tier) for frontend and backend error tracking
- Set up alerts for 5xx errors, slow responses
- Add health check endpoint monitoring (UptimeRobot free tier)

5.4 Database Maintenance

- Add database backup schedule (Render provides daily backups)
- Review and add missing database indexes
- Set up connection monitoring

Phase 6: Launch Prep (Week 6)

Priority: Critical — everything needed to go live

6.1 Legal & Compliance

- Finalize privacy policy (GDPR-compliant: data collection, cookies, third-party sharing)
- Finalize terms of service

- Cookie consent banner (if serving EU users)
- Affiliate disclosure page (FTC compliance)
- LEGO trademark disclaimer ("LEGO is a trademark of the LEGO Group, which does not sponsor this site")

6.2 Domain & DNS

- Purchase and configure production domain
- Set up SSL certificate (auto via Vercel/Render)
- Configure custom domain on Vercel (frontend) and Render (API)
- Set up email sending domain (for transactional emails)

6.3 Production Environment

- Review all environment variables for production values
- Disable debug/dev flags (ALLOW_FAKE_AUTH, AUTH_DEBUG)
- Set production SECRET_KEY (strong, unique)
- Configure CORS for production domain only
- Enable HTTPS-only redirects

6.4 SEO Final Check

- Submit sitemap to Google Search Console
- Submit to Bing Webmaster Tools
- Verify robots.txt is correct for production domain
- Test OG images and meta tags with social media debuggers
- Add structured data (JSON-LD) for set pages (Product schema)

6.5 Analytics & Monitoring

- Verify GA4 tracking on production
- Set up Google Search Console
- Create GA4 goals/conversions (signup, affiliate click, review posted)
- Dashboard for key metrics

6.6 Content & Data

- Ensure set database is fully populated and current
- Create 5-10 curated featured lists (staff picks)
- Write compelling homepage copy
- Create a few seed reviews for popular sets (from your own accounts)

Phase 7: Growth & Monetization (Post-Launch, Ongoing)

Priority: Revenue generation and user growth

7.1 Affiliate Revenue

- Apply to Amazon Associates program (need qualifying sales first)
- Apply to LEGO.com affiliate program (via Rakuten/CJ)
- Apply to Target, Walmart affiliate programs

- Track affiliate earnings and optimize placement

7.2 SEO Content Strategy

- Create themed landing pages ("Best Star Wars Sets 2026", "Sets Under $50")
- Blog/guides section for organic traffic
- User-generated content (reviews, lists) as SEO driver
- Internal linking strategy between related sets/themes

7.3 Social Media Presence

- Set up Instagram, Twitter/X, TikTok accounts
- Share new releases, retiring alerts, popular reviews
- Engage with LEGO community (Reddit r/lego, BrickLink forums)

7.4 User Growth Tactics

- Shareable collection/list pages with OG previews
- "Compare collections with friends" feature
- Referral program (future)
- Google/Facebook login (OAuth) for frictionless signup

7.5 Future Monetization Options

- Premium tier ($5/mo): advanced price alerts, price history, collection analytics, no ads
- Display ads on browse/search pages (non-intrusive, post-scale)
- Sponsored set placements (partner with LEGO stores)

Timeline Summary

Week	Dates	Focus	Deliverables
1	Mar 9-15	Security & Auth	Password reset, email verification, admin auth, rate limiting
2	Mar 16-22	Design Pass (Part 1)	Home page redesign, set detail polish, design system
3	Mar 23-29	Design Pass (Part 2) + Email	Collection UX, mobile polish, onboarding, email setup
4	Mar 30 - Apr 5	Price Data (Part 1)	MSRP from Rebrickable, Amazon PA-API setup, offers DB
5	Apr 6-12	Price Data (Part 2) + Performance	Price display, caching, Core Web Vitals, error monitoring
6	Apr 13-19	Launch Prep	Legal, domain, production config, SEO, content seeding
7	Apr 20-26	**SOFT LAUNCH**	Invite friends/family, fix bugs, gather feedback

Week	Dates	Focus	Deliverables
8	Apr 27 - May 3	Bug Fixes & Polish	Address feedback, performance tuning
9	May 4-10	**PUBLIC LAUNCH**	Submit to directories, social media, SEO push
10+	May 11+	Growth & Monetization	Affiliate programs, content, social, iterate on feedback

Soft Launch: ~April 20 | Public Launch: ~May 4 | Buffer through May 31

Cost Estimates (Monthly)

Service	Plan	Cost
Render (Backend)	Starter	$7/mo
Render (PostgreSQL)	Starter	$7/mo
Vercel (Frontend)	Hobby (free) → Pro	$0-20/mo
Domain	Annual	~$12/yr ($1/mo)
Resend (Email)	Free tier (3k emails/mo)	$0/mo
Sentry (Error monitoring)	Free tier	$0/mo
UptimeRobot (Monitoring)	Free tier	$0/mo
Google Analytics	Free	$0/mo
Total (Launch)		**~$15-35/mo**

Scale costs (1k+ users): Render Pro ($25/mo), Vercel Pro ($20/mo), Redis ($15/mo) = ~$75/mo

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Key Metrics to Track

Metric	Target (Month 1)	Target (Month 3)
Registered users	100	500
Monthly active users	50	250
Sets tracked (total across users)	1,000	10,000
Reviews written	50	300
Affiliate clicks/month	200	1,000
Email subscribers	200	1,000
Google organic visits/day	10	100

Risk Mitigation

Risk	Mitigation
Amazon PA-API access denied	Start with MSRP display only; apply after qualifying sales
Low initial user growth	Focus on SEO long-tail content; engage Reddit/LEGO communities
Rebrickable API downtime	Local cache with 24h fallback; DB stores all set data
Cost overrun at scale	Free tiers cover first ~1k users; monetize before scaling
LEGO trademark issues	Clear disclaimer on every page; no official branding confusion
Data freshness	Automated daily refresh via cron; retirement date monitoring

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Success Criteria for Launch

- [ ] All auth flows work (register, login, logout, password reset, email verify)
- [ ] Search returns relevant results with working filters
- [ ] Users can build and share collections/lists
- [ ] Reviews and ratings work end-to-end
- [ ] At least MSRP prices displayed on set pages
- [ ] Affiliate links track clicks correctly
- [ ] Email signups captured and welcome email sent
- [ ] All pages render correctly on mobile
- [ ] Core Web Vitals pass (LCP < 2.5s, CLS < 0.1)
- [ ] No console errors on any page
- [ ] Production environment secured (no dev flags, proper CORS)
- [ ] Legal pages complete and linked
- [ ] Sitemap submitted to Google Search Console

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